Shiyun Xiong

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Education

Harbin Institute of Technology **Weihai, China**
Bachelor of Engineering in Software Engineering *Aug. 2021 – Jun. 2025*

- Weighted Average: 90.0/100
- Main course: Linear Algebra(92), Calculus(92), Algorithm Analysis and Design(97), Probability theory and mathematical statistics(90), Introduction to Artificial Intelligence(92), Data Structure(94), Database System(97), Operating Systems(94)

Research Experiences

Adaptive Architecture for Embodied-Sensory Omni-Modal LLMs **Advisor: Prof. Xiangyu Yue**
Multimedia Laboratory, The Chinese University of Hong Kong (MMLab, CUHK) *Jan. 2026 – Present*

- This project aims to develope an adaptive omni-modal LLM architecture for embodied sensory understanding, extending multimodal reasoning from conventional vision-language inputs to heterogeneous sensory modalities such as depth, IMU, pointcloud, thermal, and tactile signals.
- My contributions: Developed the adaptive omni-modal architecture with sensory-specific projection, caption-supervised representation alignment, and modality-aware MoE specialization; curated heterogeneous embodied sensory data; and conducted multi-stage training and comprehensive evaluations.

Autonomous Benchmark Construction Agent for Multimodal Model Evaluation **Advisor: Prof. Xiangyu Yue**
Multimedia Laboratory, The Chinese University of Hong Kong (MMLab, CUHK) *Sep. 2025 – Jan. 2026*

- This project aims to develope the Benchmark Agent, an autonomous agentic system that transforms open-ended evaluation intents into customized multimodal benchmarks through long-horizon planning, tool-augmented reasoning, and self-reflective refinement.
- My contributions: Developed the closed-loop agentic workflow that integrates intent-aware planning, grounded reasoning, constraint-driven execution, and self-verification, enabling scalable generation of reliable benchmarks with controllable evaluation objectives and minimal human supervision.

Self-Evolving Intelligent Agents in Visual Web Environments **July. 2024 – Feb. 2025**

- This project aims to enhance agent interaction within various visual digital platforms using a proprietary reward model. The goal is to enable agents to adapt based on user interactions and environmental feedback.
- My contributions: Developed the agent framework where agents interact with the different environments and constructed training data for different reward models based on existing data features. Conducted experiments in both static and dynamic environments to evaluate the effectiveness of the framework and its performance in diverse scenarios.

Multi-Agent Framework for Web Services with Fuzzy Requirements **Advisor: Prof. Zhiying Tu**
Research Center of Intelligent Computing for Enterprises & Services (ICES, HIT) *Mar 2024 – July.2024*

- This project implemented H-Tower, the first LLM-based multi-agent framework simulating multi-turn interactions between humans and websites. It achieves emotion-aware elicitation of complex requirements and mass-produces training data for this task.
- My contributions: Developed the system framework and designed multi-agent collaboration mechanisms to resolve fuzzy user requirements in conversational settings, improving the efficiency and completeness of user need elicitation.

A Benchmark for Universal Instruction Following in Realistic Web Services Navigation **Advisor: Prof. Dianhui Chu**
Research Center of Intelligent Computing for Enterprises & Services (ICES, HIT) *Nov. 2023 – Mar. 2024*

- This project constructed the first Chinese multi-modal benchmark and dataset to evaluate web service navigation tasks and devised an efficient multi-modal framework to evaluate performance on the dataset.
- My contributions: Developed a multi-modal framework integrating slot tree maintenance, instruction parsing, and web execution, achieving a 68.61% success rate in automatic web navigation. Proposed using object detection to filter noise from web pages, converting visual information into language data and fine-tuned a large language model for better slot tree utilization.

Publications

Shiyun Xiong, Dongming Wu, Peiwen Sun, Yuang Ai, Bokang Yang, Wencheng Han, Xiao-Hui Li, Xiangyu Yue **"Benchmark Everything Everywhere All at Once."** *Under Review*

Bolin Zhang, Dianhui Chu (*Member, IEEE*), **Shiyun Xiong**, Qianjun Gong, and Zhiying Tu (*Member, IEEE*) **"User-Oriented Fuzzy Requirement Clarification for Services Delivery in the Metaverse."** *IEEE Transactions on Services Computing*

Zhiyuan Hu, **Shiyun Xiong**, Yifan Zhang, See-Kiong Ng, Anh Tuan Luu, Bo An, Shuicheng YAN, Bryan Hooi **"Guiding VLM Agents with Process Rewards at Inference Time for GUI Navigation."** *Preprint*

Bolin Zhang, **Shiyun Xiong**, Dianbo Sui, Yunzhe Xu, Zhiying Tu, and Dianhui Chu. 2024. **"RealWeb: A Benchmark for Universal Instruction Following in Realistic Web Services Navigation."** In *2024 IEEE International Conference on Web Services*

Service and Skills

Reviewer: ICLR 2025 Workshop LLM Reason and Plan, ACL ARR 2025 February

Technical Skills and Tools: Python, C++, C, Java, Matlab, Java, Pytorch, SQL, Selenium, Android Studio, etc.

Languages: Chinese Mandarin (naive), English (CET 4: 613, CET 6: 564)